<u>Descriptive Statistics Assignment</u>

Q1: Understanding Central Tendency (Easy)
A bakery tracks the daily sales of muffins (in dozens) over a week: [10, 12, 11, 15, 14, 13, 12]. What is the most representative value of their weekly sales, and why?

Ans- The most representative value of the bakery's weekly muffin sales is approximately 12.4 dozen, which is the mean. It is calculated by adding all daily sales (87) and dividing by 7 days. The mean represents the average daily sales and considers all data values. The median and mode are both 12, which also supports that the typical daily sales are around 12 dozen. Therefore, the bakery usually sells about 12 to 13 dozen of muffins per day.

Q2: Mean in Real Life (Easy)
A teacher records the marks of her students in a short quiz: [12, 15, 14, 16, 18, 20, 19]. What is the mean score, and what does it tell us about the class's performance?

Ans- The mean score of the class is 16.29, calculated by adding all the marks (114) and dividing by the total number of students (7). This means that the average performance of the class is approximately 16 marks. It shows that the students performed well overall, as most of the scores are above 15. The mean gives a good summary of the class's general performance.

Q3: Mode in Real Life (Easy)
A store records the shoe sizes sold in one day: [7, 8, 9, 8, 8, 10, 7, 9]. What is the mode, and why is this information useful for the store manager?

Ans- The mode of the shoe sizes sold is 8, because it appears three times, which is more than any other size. This information is useful for the store manager because it shows which shoe size is most in demand. The manager can use this information to stock more size 8 shoes to meet customer demand and improve sales.

Q4: Median in Real Life (Medium)
A car dealer notes the prices of used cars: [$8,000, $9,500, $10,200, $11,000, $50,000]. Why is the median a better measure than the mean in this case? Calculate the median.

Ans- The Median of the given data is $10,200 The median is a better measure than the mean because the dataset contains an extreme value ($50,000), which increases the mean significantly. The median is not affected by outliers and better represents the typical car price. Since there are five values, the median is the middle value, which is $10,200. Therefore, the median gives a more accurate picture of the usual used car price in this dataset.

Q5: Dispersion Introduction (Medium)
A student times how long it takes to finish a puzzle each day: [25, 30, 27, 35, 40]. What does the range tell us about the variation in the student's puzzle-solving time?

Ans- As in the question the data is not properly mentioned hence, we are assuming the given data in minutes. So, the range is calculated by subtracting the minimum value from the maximum value. The maximum puzzle-solving time is 40 minutes and the minimum time is 25 minutes. Therefore, the range is 15 minutes. This means the student's puzzle-solving time varies by 15 minutes from the fastest to the slowest day. The range shows the amount of variation in the student's performance over the five days

Q6: Range in Action (Medium)
A farmer records the weekly weight of harvested apples (kg): [100, 105, 98, 110, 120]. Find the range. How can this help the farmer in planning his packaging?

Ans- The range is calculated by subtracting the minimum value from the maximum value. The maximum weight is 120 kg and the minimum weight is 98 kg. Therefore, the range is 22 kg. This means the weekly harvest varies by 22 kg. This information helps the farmer plan packaging materials properly, ensuring enough boxes are available during high-production weeks and avoiding waste during low-production weeks.

Q7: Variance for Decision-Making (Medium)
Two delivery companies track delivery delays (in minutes). Company A: variance = 6 Company B: variance = 15Which company is more consistent, and why?

Ans- Company A is more consistent because it has a lower variance (6) compared to Company B (15). Variance measures how much the delivery delays vary from the average. A lower variance means the delays are more closely grouped around the mean, indicating more stable and reliable performance. Therefore, Company A shows greater consistency in delivery times.

Q8: Standard Deviation in Context (Hard)
A finance student compares the daily price fluctuations of two cryptocurrencies. Coin A: standard deviation = $30 Coin B: standard deviation = $120 Which coin is riskier to invest in, and why?

Ans- Coin B is riskier to invest in because it has a higher standard deviation ($120) compared to Coin A ($30). Standard deviation measures how much prices fluctuate from the average. A higher standard deviation indicates greater volatility and larger price movements. Since Coin B's price varies more significantly, it carries higher investment risk.

Q9: Combining Measures (Hard)
A family records their monthly electricity usage (in kWh): [400, 420, 390, 450, 410]. Find the mean and standard deviation. What do these values together tell you about the family's energy use pattern?

Ans- The mean monthly electricity usage is 414 kWh, calculated by dividing the total usage (2070 kWh) by 5 months. The standard deviation is approximately 23 kWh, which measures how much the monthly usage varies from the average. Together, these values show that the family's energy consumption is relatively consistent, with only moderate fluctuations around the average usage.

Q10: Practical Application (Hard)
A basketball player's points in 8 games are recorded: [15, 18, 20, 22, 25, 17, 19, 21]. Find the mean, median, mode, range, and standard deviation. What insights can these measures provide about the player's scoring performance?

Ans- The mean score is 19.63 points, calculated by dividing the total points (157) by 8 games. The median is 19.5 points, which is the average of the 4th and 5th values after arranging the data. There is no mode because no score repeats. The range is 10 points (25 − 15), showing the difference between the highest and lowest score. The standard deviation is approximately 3.16 points, indicating that the player's scores are closely grouped around the mean. Overall, these measures show that the player performs consistently and scores around 20 points per game with moderate variation.

Formulas Used in Assignments Are:

$\text{Mean} = n\sum X$

Median=Middle Value

Median=2Value1+Value/2

Range=Maximum−Minimum

$S\,2 = \sum(X-\bar{X})2 \,/\, n-1$

$S = \sqrt{\ } \sum(X-\bar{X})2 \,/\, n-1$